Cluster Analysis

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```
data<- read.csv('/Users/adithyaprahasith/Downloads/Pharmaceuticals.csv')
head(data)
```

```
##   Symbol                 Name Market_Cap Beta PE_Ratio  ROE  ROA
Asset_Turnover
## 1    ABT Abbott Laboratories      68.44 0.32     24.7 26.4 11.8
0.7
## 2    AGN      Allergan, Inc.       7.58 0.41     82.5 12.9  5.5
0.9
## 3    AHM        Amersham plc       6.30 0.46     20.7 14.9  7.8
0.9
## 4    AZN     AstraZeneca PLC      67.63 0.52     21.5 27.4 15.4
0.9
## 5    AVE             Aventis      47.16 0.32     20.1 21.8  7.5
0.6
## 6    BAY            Bayer AG      16.90 1.11     27.9  3.9  1.4
0.6
##   Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location
Exchange
## 1     0.42       7.54              16.1          Moderate Buy       US
NYSE
## 2     0.60       9.16               5.5          Moderate Buy   CANADA
NYSE
## 3     0.27       7.05              11.2            Strong Buy       UK
NYSE
## 4     0.00      15.00              18.0         Moderate Sell       UK
NYSE
## 5     0.34      26.81              12.9          Moderate Buy   FRANCE
NYSE
## 6     0.00      -3.17               2.6                  Hold  GERMANY
NYSE
```

```
str(data)
```

```
## 'data.frame':    21 obs. of  14 variables:
##  $ Symbol                : chr  "ABT" "AGN" "AHM" "AZN" ...
##  $ Name                  : chr  "Abbott Laboratories" "Allergan, Inc."
"Amersham plc" "AstraZeneca PLC" ...
##  $ Market_Cap            : num  68.44 7.58 6.3 67.63 47.16 ...
##  $ Beta                  : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08
0.18 ...
##  $ PE_Ratio              : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6
27.9 ...
##  $ ROE                   : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1
```

```
31 ...
##  $ ROA                  : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5
...
##  $ Asset_Turnover       : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
##  $ Leverage             : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53
...
##  $ Rev_Growth           : num  7.54 9.16 7.05 15 26.81 ...
##  $ Net_Profit_Margin    : num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3
23.4 ...
##  $ Median_Recommendation: chr  "Moderate Buy" "Moderate Buy" "Strong Buy"
"Moderate Sell" ...
##  $ Location             : chr  "US" "CANADA" "UK" "UK" ...
##  $ Exchange             : chr  "NYSE" "NYSE" "NYSE" "NYSE" ...
```

```
summary(data)
```

```
##     Symbol              Name             Market_Cap          Beta
##  Length:21          Length:21          Min.   :  0.41   Min.   :0.1800
##  Class :character   Class :character   1st Qu.:  6.30   1st Qu.:0.3500
##  Mode  :character   Mode  :character   Median : 48.19   Median :0.4600
##                                        Mean   : 57.65   Mean   :0.5257
##                                        3rd Qu.: 73.84   3rd Qu.:0.6500
##                                        Max.   :199.47   Max.   :1.1100
##     PE_Ratio          ROE             ROA          Asset_Turnover    Leverage
##  Min.   : 3.60   Min.   : 3.9   Min.   : 1.40   Min.   :0.3    Min.
:0.0000
##  1st Qu.:18.90   1st Qu.:14.9   1st Qu.: 5.70   1st Qu.:0.6    1st
Qu.:0.1600
##  Median :21.50   Median :22.6   Median :11.20   Median :0.6    Median
:0.3400
##  Mean   :25.46   Mean   :25.8   Mean   :10.51   Mean   :0.7    Mean
:0.5857
##  3rd Qu.:27.90   3rd Qu.:31.0   3rd Qu.:15.00   3rd Qu.:0.9    3rd
Qu.:0.6000
##  Max.   :82.50   Max.   :62.9   Max.   :20.30   Max.   :1.1    Max.
:3.5100
##    Rev_Growth    Net_Profit_Margin Median_Recommendation   Location
##  Min.   :-3.17   Min.   : 2.6      Length:21             Length:21
##  1st Qu.: 6.38   1st Qu.:11.2      Class :character      Class :character
##  Median : 9.37   Median :16.1      Mode  :character      Mode  :character
##  Mean   :13.37   Mean   :15.7
##  3rd Qu.:21.87   3rd Qu.:21.1
##  Max.   :34.21   Max.   :25.5
##    Exchange
##  Length:21
##  Class :character
##  Mode  :character
##
##
##
```

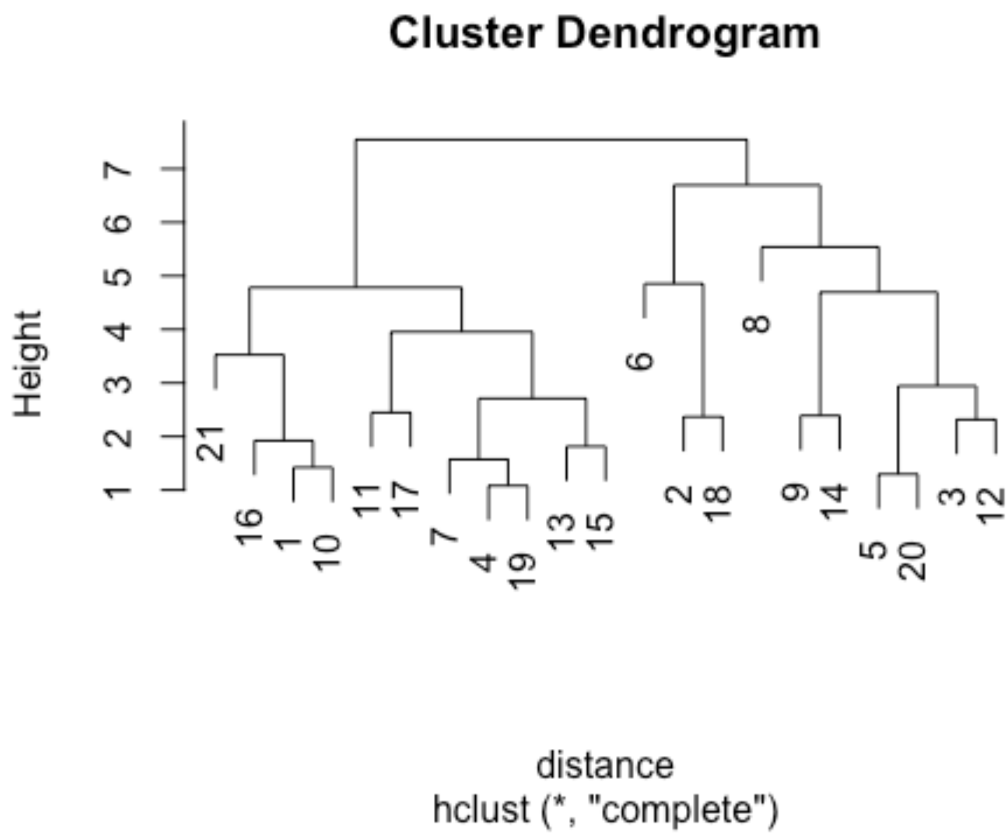
```r
# Normalize
z <- data[,-c(1,2,12,13,14)]
means <- apply(z,2,mean)
sds <- apply(z,2,sd)
nor <- scale(z,center=means,scale=sds)
head(nor)
```

```
##      Market_Cap        Beta    PE_Ratio         ROE        ROA
Asset_Turnover
## [1,]  0.1840960 -0.80125356 -0.04671323  0.04009035  0.2416121
0.0000000
## [2,] -0.8544181 -0.45070513  3.49706911 -0.85483986 -0.9422871
0.9225312
## [3,] -0.8762600 -0.25595600 -0.29195768 -0.72225761 -0.5100700
0.9225312
## [4,]  0.1702742 -0.02225704 -0.24290879  0.10638147  0.9181259
0.9225312
## [5,] -0.1790256 -0.80125356 -0.32874435 -0.26484883 -0.5664461     -
0.4612656
## [6,] -0.6953818  2.27578267  0.14948233 -1.45146000 -1.7127612     -
0.4612656
##        Leverage Rev_Growth Net_Profit_Margin
## [1,] -0.2120979 -0.5277675        0.06168225
## [2,]  0.0182843 -0.3811391       -1.55366706
## [3,] -0.4040831 -0.5721181       -0.68503583
## [4,] -0.7496565  0.1474473        0.35122600
## [5,] -0.3144900  1.2163867       -0.42597037
## [6,] -0.7496565 -1.4971443       -1.99560225
```

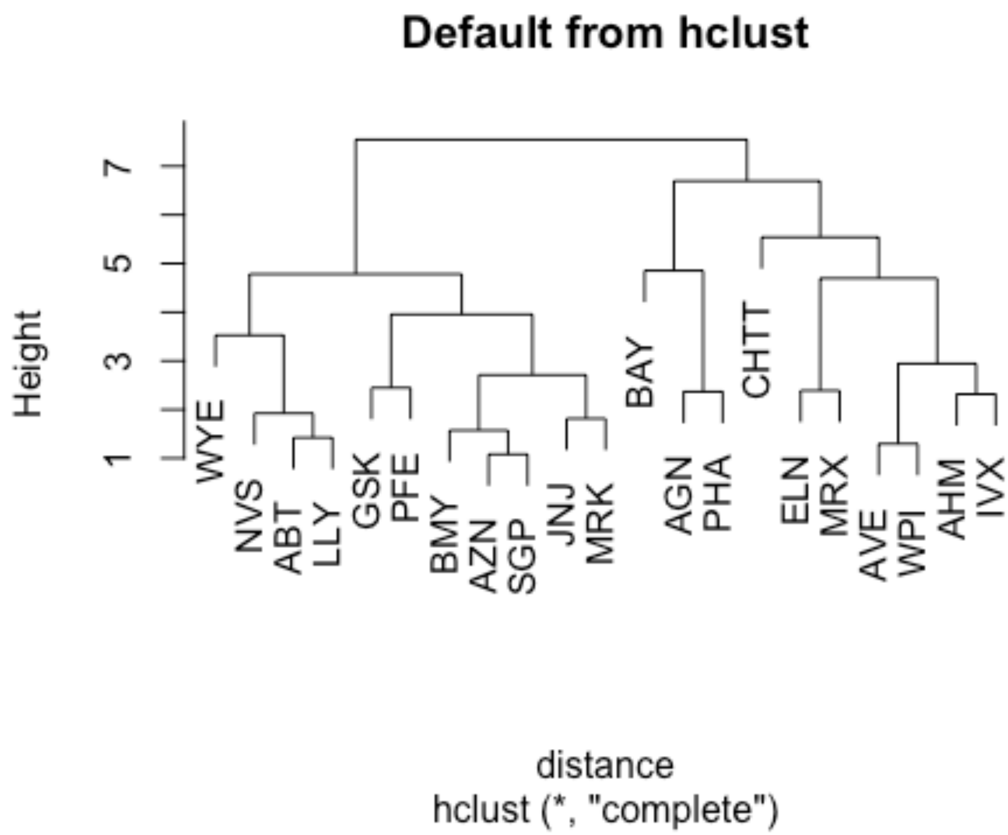
```r
# Calculate distance matrix
distance = dist(nor)
distance
```

```
##           1        2        3        4        5        6        7        8
## 2  4.415575
## 3  2.018793 3.945745
## 4  1.669541 4.909566 2.364249
## 5  2.111983 4.642699 2.487172 2.632282
## 6  4.690231 4.853901 3.636353 5.065563 4.764654
## 7  1.805543 5.419487 2.600986 1.572582 3.400602 5.273023
## 8  5.020726 5.612226 4.760341 5.719174 5.096246 4.969438 5.287400
## 9  4.901141 6.695261 4.695844 4.974521 3.748778 4.608660 5.378092 4.675606
## 10 1.422680 5.140253 3.238353 2.405951 2.910766 5.804419 2.189107 5.657801
## 11 3.689906 6.747789 4.904614 2.957494 4.476690 7.546154 3.099023 7.080175
## 12 2.624729 4.470028 2.316548 3.282195 2.386850 3.658011 3.279927 2.951511
## 13 2.333874 5.317942 3.593764 1.958326 3.640773 5.724303 2.511309 6.310233
## 14 3.920297 5.479080 4.120549 4.269231 2.927258 4.848442 4.734766 4.786213
## 15 2.680733 5.443918 3.361981 1.859280 3.472410 5.918477 2.432281 6.101541
## 16 1.922731 5.468844 3.331743 3.056196 3.330879 5.331004 2.866126 6.063738
## 17 3.887235 6.906828 5.268858 3.109413 4.495242 7.163993 3.666674 7.180257
```

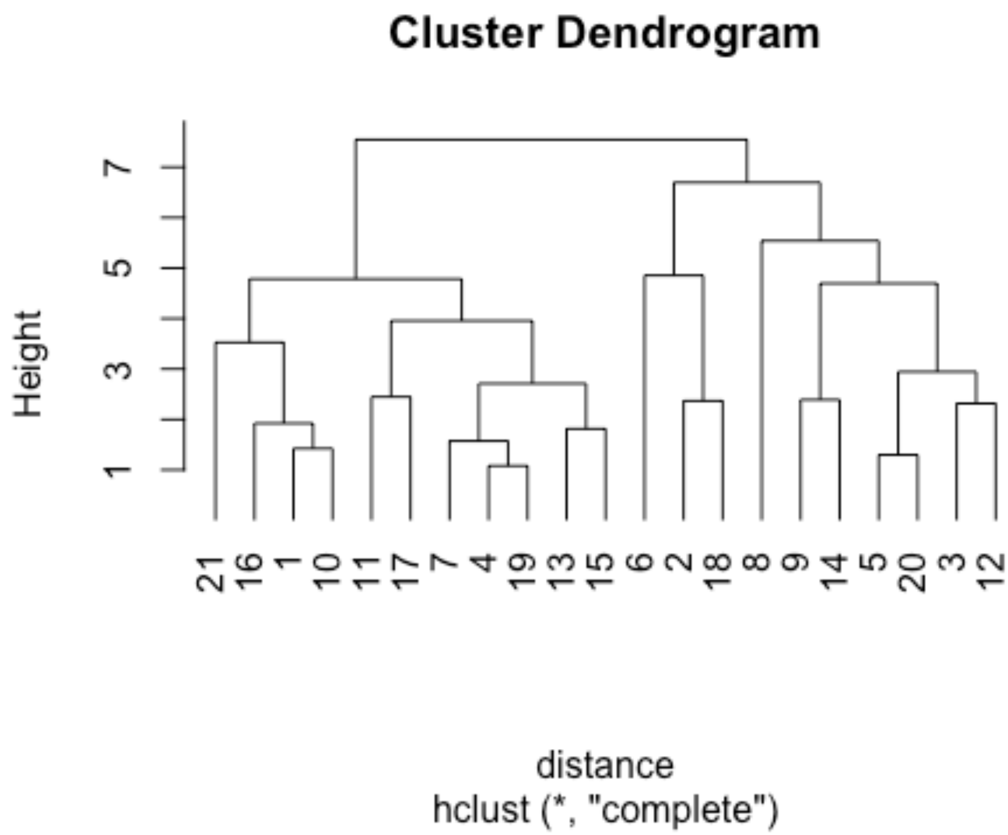
```
## 18 2.908982 2.367912 2.925627 3.715808 2.718441 3.955926 4.408645 5.000709
## 19 1.312599 4.725384 1.704709 1.080519 2.464855 4.426418 1.478433 5.346513
## 20 2.882610 5.007086 2.943946 3.414127 1.296549 5.055769 4.116074 5.540296
## 21 3.038549 6.446458 4.185594 3.324966 4.254562 5.954379 2.269808 5.127981
##           9       10       11       12       13       14       15       16
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10 5.554227
## 11 6.731204 3.631174
## 12 3.115283 3.537378 5.276601
## 13 6.070533 2.722434 2.988672 4.354581
## 14 2.389723 4.191466 6.187185 2.825394 5.306512
## 15 5.921987 3.380695 2.218040 4.164267 1.814184 5.532520
## 16 5.732322 1.577953 4.783039 3.899915 3.083678 4.478040 4.112418
## 17 6.123133 3.783136 2.447177 5.356598 2.447341 5.518379 2.831329 4.536250
## 18 5.007721 3.754900 5.773960 3.073579 4.112432 3.827019 4.448933 3.884035
## 19 4.665611 2.205815 3.780283 2.763476 2.604437 3.907501 2.710607 2.542763
## 20 3.756437 3.412378 5.437193 2.857109 4.591764 2.653341 4.569336 3.626404
## 21 5.312455 2.747839 3.670720 3.719962 3.858028 4.709401 3.935039 3.525940
##          17       18       19       20
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18 5.587119
## 19 3.955078 3.449579
## 20 5.403128 3.172178 3.026610
## 21 4.026095 5.286507 3.145472 4.922945

# Hierarchical agglomerative clustering
mydata.hclust = hclust(distance)
plot(mydata.hclust)
```



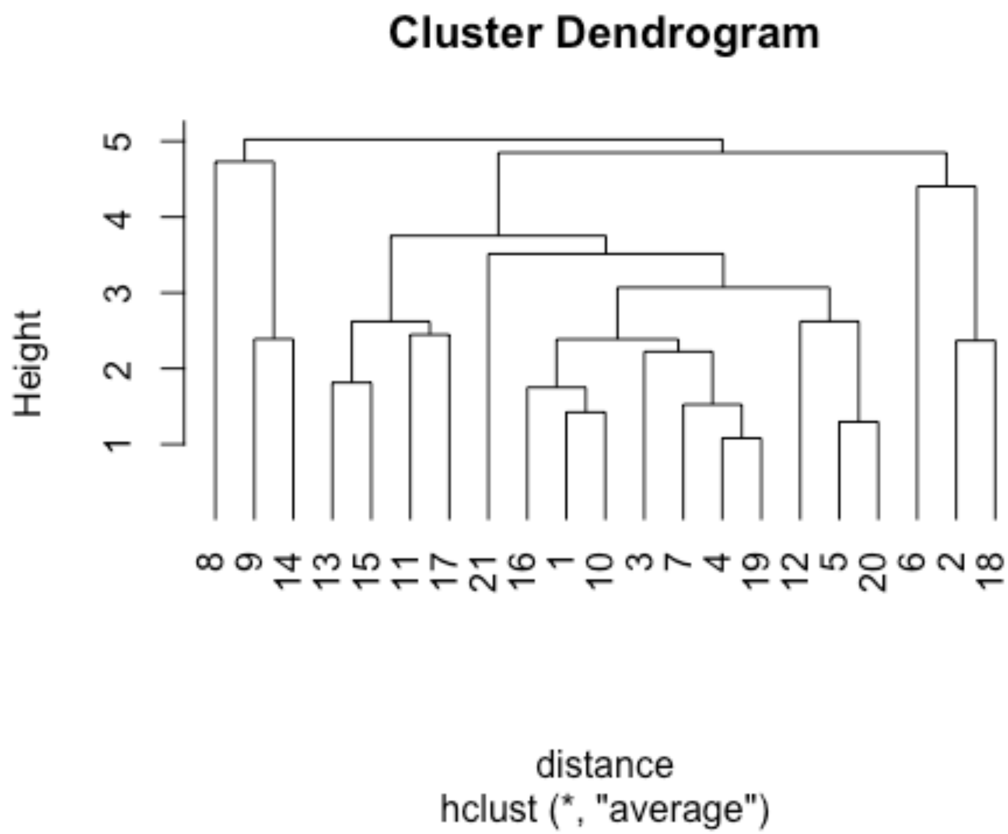

```
plot(mydata.hclust,labels=data$Symbol,main='Default from hclust')
```




```
plot(mydata.hclust,hang=-1)
```

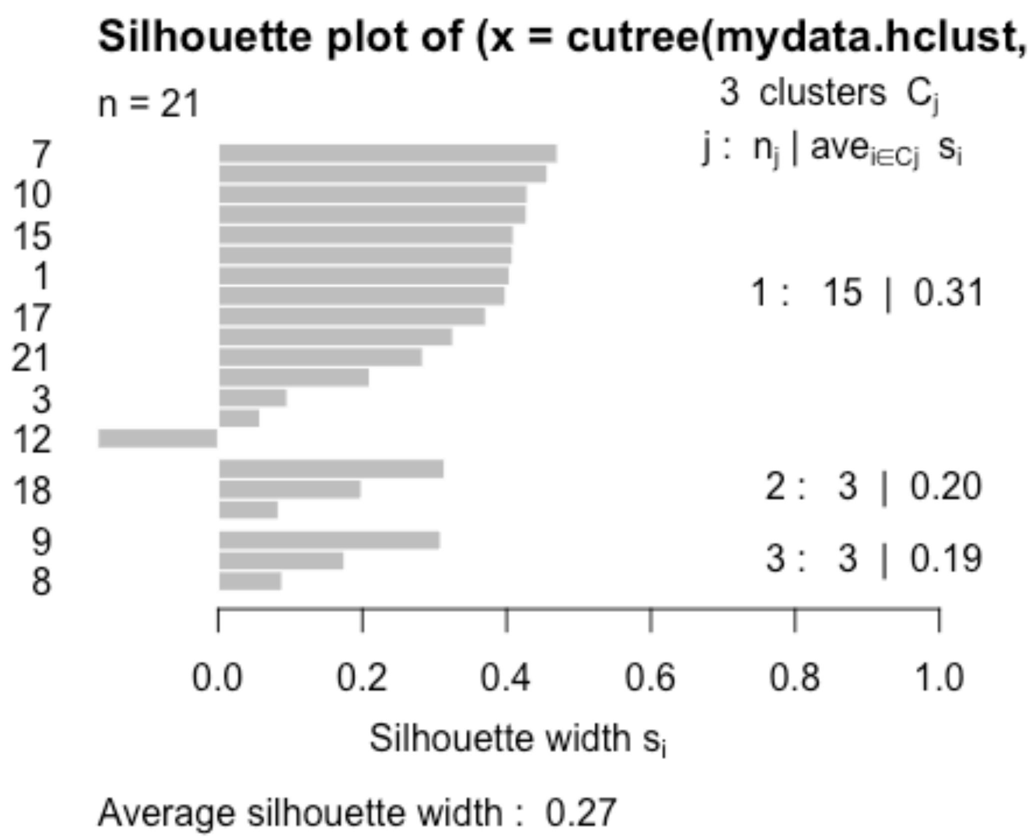



```r
# Hierarchical agglomerative clustering using "average" linkage
mydata.hclust<-hclust(distance,method="average")
plot(mydata.hclust,hang=-1)
```

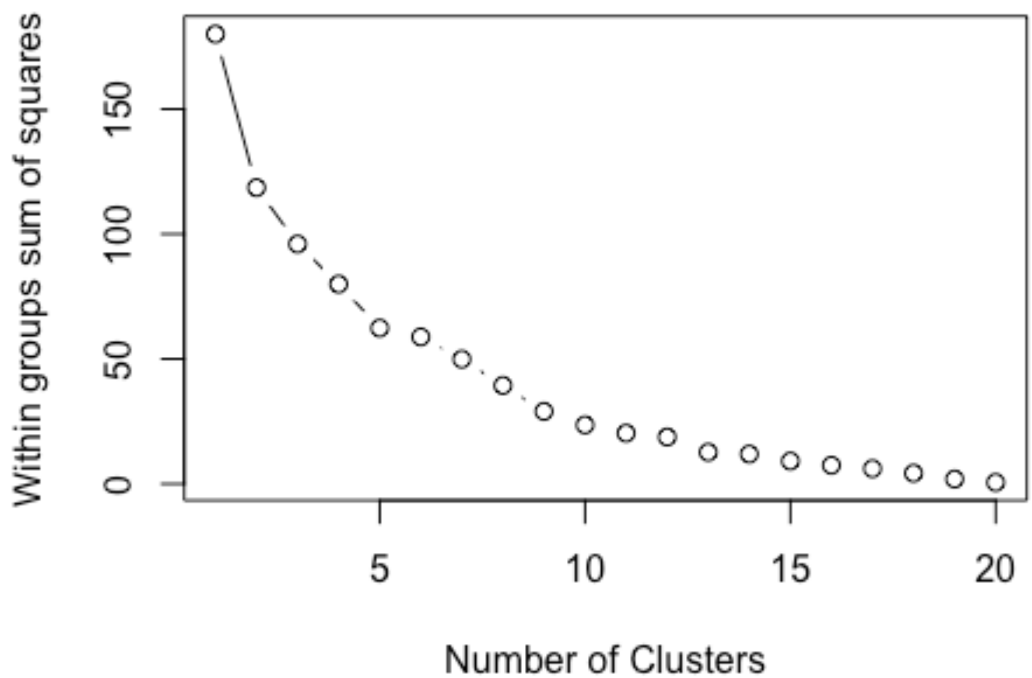



```
# Cluster membership
member = cutree(mydata.hclust,3)
table(member)

## member
##  1  2  3
## 15  3  3

# Characterizing clusters
aggregate(nor,list(member),mean)

##   Group.1 Market_Cap       Beta   PE_Ratio        ROE        ROA
Asset_Turnover
## 1       1  0.2989594 -0.3754021 -0.2956363  0.3273519  0.4470718
2.767594e-01
## 2       2 -0.5246281  0.4451409  1.8498439 -1.0404550 -1.1865838
1.480297e-16
## 3       3 -0.9701688  1.4318698 -0.3716621 -0.5963045 -1.0487754  -
1.383797e+00
##     Leverage  Rev_Growth Net_Profit_Margin
## 1 -0.2513482 -0.07074541         0.3725608
## 2 -0.3443544 -0.57694536        -1.6095439
## 3  1.6010956  0.93067243        -0.2532601
```

```
aggregate(data[,-c(1,2,12,13,14)],list(member),mean)

##   Group.1 Market_Cap      Beta PE_Ratio      ROE       ROA Asset_Turnover
## 1       1 75.1713333 0.4293333 20.64000 30.73333 12.893333           0.76
## 2       2 26.9066667 0.6400000 55.63333 10.10000  4.200000           0.70
## 3       3  0.7966667 0.8933333 19.40000 16.80000  4.933333           0.40
##    Leverage Rev_Growth Net_Profit_Margin
## 1 0.3893333  12.589333         18.140000
## 2 0.3166667   6.996667          5.133333
## 3 1.8366667  23.653333         14.033333

# Silhouette Plot
library(cluster)
plot(silhouette(cutree(mydata.hclust,3), distance))
```




```
# Scree Plot
wss <- (nrow(nor)-1)*sum(apply(nor,2,var))
for (i in 2:20) wss[i] <- sum(kmeans(nor, centers=i)$withinss)
plot(1:20, wss, type="b", xlab="Number of Clusters", ylab="Within groups sum
of squares")
```



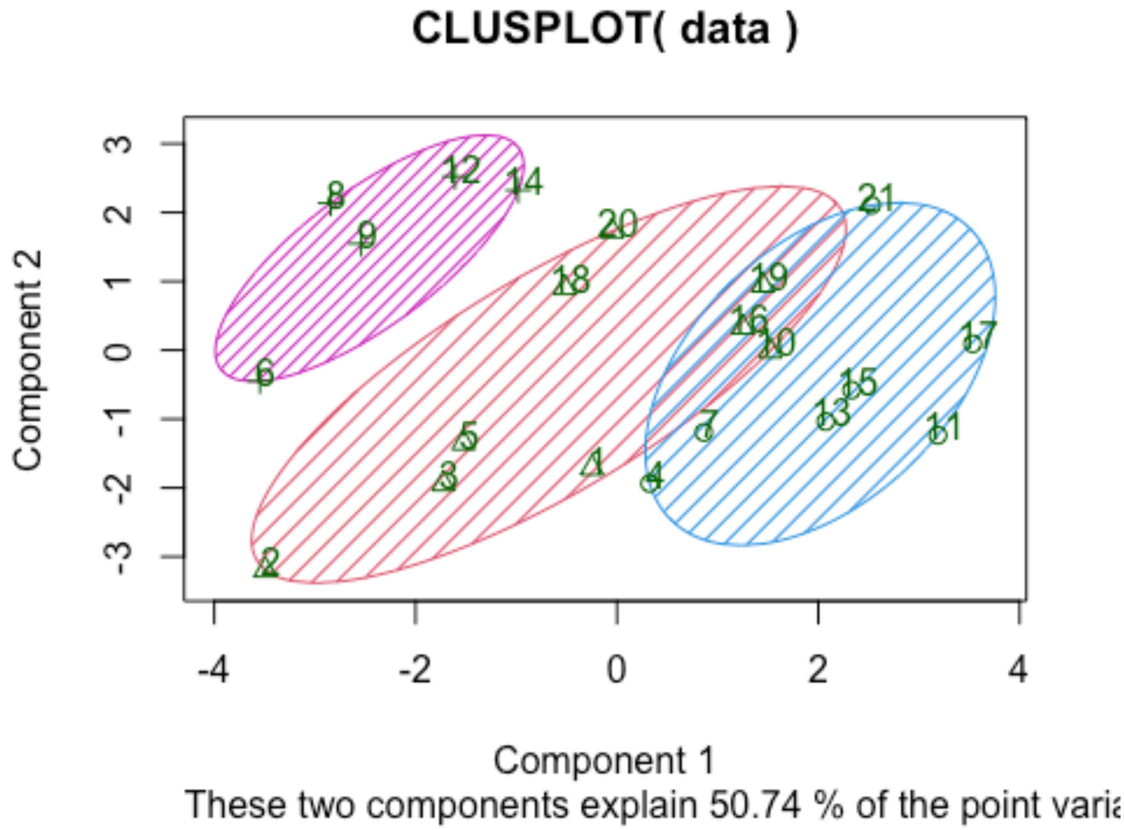

```
# K-means clustering
set.seed(123)
kc<-kmeans(nor,3)
kc

## K-means clustering with 3 clusters of sizes 7, 9, 5
##
## Cluster means:
##    Market_Cap        Beta   PE_Ratio        ROE        ROA Asset_Turnover
## 1  0.9547543 -0.06120687 -0.3576482  1.0818081  1.1033619      0.8566361
## 2 -0.2375550 -0.73633718  0.4233386 -0.4489909 -0.2407172     -0.1025035
## 3 -0.9090570  1.41109654 -0.2613021 -0.7063477 -1.1114156     -1.0147843
##     Leverage  Rev_Growth Net_Profit_Margin
## 1 -0.2797499 -0.01818848         0.7082574
## 2 -0.3557313 -0.13595383        -0.1652117
## 3  1.0319661  0.27018076        -0.6941793
##
## Clustering vector:
##  [1] 2 2 2 1 2 3 1 3 3 2 1 3 1 3 1 2 1 2 2 2 1
##
## Within cluster sum of squares by cluster:
## [1] 25.26414 42.25037 31.94053
##  (between_SS / total_SS =  44.7 %)
```

```
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"
"tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
companies_cluster <- data.frame(Company=data$Symbol, Cluster_ = kc$cluster)
companies_cluster
```

```
##    Company Cluster_
## 1      ABT        2
## 2      AGN        2
## 3      AHM        2
## 4      AZN        1
## 5      AVE        2
## 6      BAY        3
## 7      BMY        1
## 8     CHTT        3
## 9      ELN        3
## 10     LLY        2
## 11     GSK        1
## 12     IVX        3
## 13     JNJ        1
## 14     MRX        3
## 15     MRK        1
## 16     NVS        2
## 17     PFE        1
## 18     PHA        2
## 19     SGP        2
## 20     WPI        2
## 21     WYE        1
```

```
clusplot(data, kc$cluster,
         color = TRUE,
         shade = TRUE,
         labels = 3,
         lines = 0)
```



1. Use only the quantitative variables (a)-(i) to cluster the 21 firms. Provide final result about number of clusters chosen and cluster memberships

   Number of clusters=3, method: elbow method

   Cluster sizes are cluster 1: 7, cluster 2: 9, cluster3: 5

   Cluster memberships:

```
> companies_cluster <- data.frame(Company=data$Symbol, Cluster_ = kc$cluster)
> companies_cluster
   Company Cluster_
1      ABT        2
2      AGN        2
3      AHM        2
4      AZN        1
5      AVE        2
6      BAY        3
7      BMY        1
8     CHTT        3
9      ELN        3
10     LLY        2
11     GSK        1
12     IVX        3
13     JNJ        1
14     MRX        3
15     MRK        1
16     NVS        2
17     PFE        1
18     PHA        2
19     SGP        2
20     WPI        2
21     WYE        1
```

Companies in Cluster 1: AZN, BMY, GSK, JNJ, MRK, PFE, WYE

Companies in Cluster 2: ABT, AGN, AHM, AVE, LLY, NVS, PHA, SGP, WPI

Companies in Cluster 3: BAY, CHTT, ELN, IVX, MRX

2. Justify the choices made in conducting the cluster analysis (in the previous question) such as weights accorded different variables, the specific clustering algorithm/s used, the number of clusters formed, etc

   The very first choice in conducting the cluster analysis is to consider only quantitative variables and this process used a standardized set of variables giving equal importance to the variables, next to select k value in the k-means clustering I have considered using elbow method to choose the optimal number of clusters, from the elbow plot, it is clear that within-cluster sum of squares appears to slow down after 3 clusters which indicates that k=3 captures the most of the variance without overfitting and the sizes of the three clusters are well balanced.

3. Interpret the clusters with respect to the quantitative variables that were used in forming the clusters.

   Cluster 1: the companies in this cluster are companies with higher market cap and net profit margin, these companies have the higher return on equity as well.

   Cluster 2: these are companies with mid range market cap and have the highest price to earnings ratio, and highest revenue growth.

   Cluster 3: these companies indicate a very low in market cap and in net profit margin, but are highest in leverage while roa and roe being the lowest.

4. Provide an appropriate name for each cluster using key variables in the dataset.

   Cluster 1: “Class 1” or “High Performing Companies” : these companies perform strong in the market cap and the net profit margin, while also maintaining the companies with highest ROE and ROA.

   Cluster 2: “Class 2” or “Mid Performing Companies”: while these companies have the mid-size market cap and profit margin, these companies outperform in the price to earnings ratio, and in the revenue growth, but ROA not standing out.

   Cluster 3: “Class 3” or “low Performing Companies”: these companies have the lowest market performance in profit margin and market cap, but identifies in the higher leverage and ROA, ROE being the lowest.